Notes: Livdan, Sapriza, and Zhang (JF, 2009)
Financially Constrained Stock Returns

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1 Big picture

This note summarizes Livdan, Sapriza, and Zhang (2009), which builds a quantitative investment-based asset pricing model with retained earnings, debt, costly equity issuance, and collateral constraints in order to study how financial constraints affect firms' systematic risk and expected stock returns.

The paper addresses a tension in the empirical literature. Lamont, Polk, and Saa-Requejo (2001) find that more financially constrained firms, measured with the Kaplan-Zingales index, earn lower average returns, while Whited and Wu (2006), using a different constraints index, find a positive but statistically weak relation. Livdan, Sapriza, and Zhang argue that this evidence is difficult to interpret without a model that explicitly maps financial constraints into risk and expected returns.

The core economic idea is the **inflexibility mechanism**. Firms normally use investment to smooth payouts after aggregate shocks: when aggregate productivity is high, firms invest more, so dividends do not rise one-for-one with cash flows. Financial constraints weaken this smoothing channel. If collateral constraints bind, firms cannot fully finance desired investment. Their payouts therefore covary more with aggregate shocks, making their equity riskier and expected returns higher.

2 Incremental contribution of the paper

1. **Integrates corporate financing frictions into investment-based asset pricing.** The paper extends the Zhang (2005) investment-based asset pricing framework by adding debt dynamics, retained earnings, costly equity, and collateral constraints, drawing on Hennessy and Whited (2005).

2. **Provides a structural interpretation of financial constraints risk.** The true measure of financial constraints in the model is the shadow price of new debt, the Lagrange multiplier on the collateral constraint. This gives a clean object for studying whether constrained firms should earn higher expected returns.

3. **Separates financial constraints from financial distress.** The paper emphasizes that constraints mean inability to fund all desired investments, whereas distress means inability to cover current obligations. The model focuses on collateral constraints rather than default risk.

4. **Explains why the constraints-return relation weakens after controlling for size.** In simulations, univariate sorts on the shadow price show a positive return spread, but size and book-to-market largely absorb the marginal predictive power of constraints because all these variables are jointly determined by the same underlying state variables.

5. **Evaluates empirical constraints indexes inside a known model.** Since the true shadow price is observable in the simulation, the paper can compare the Kaplan-Zingales and Whited-Wu indexes as noisy proxies for true constraints.

6. **Introduces a convex leverage-return channel.** Beyond the textbook leverage effect, market leverage also increases asset risk by reducing investment flexibility, so the leverage-return relation becomes convex rather than merely linear.

3 Model environment

3.1 Technology and productivity shocks

Firm j produces output

$$y_{jt} = e^{x_t + z_{jt}} k_{jt}^{\alpha},$$

where x_t is aggregate productivity, z_{jt} is firm-specific productivity, and k_{jt} is capital. The aggregate shock is the source of systematic risk; the firm-specific shock generates cross-sectional heterogeneity.

Operating profit is

$$\pi(k_{jt}, z_{jt}, x_t) = e^{x_t + z_{jt}} k_{jt}^{\alpha} - f.$$

The fixed operating cost f generates operating leverage, which helps connect the model to cross-sectional return patterns.

3.2 Investment and adjustment costs

Capital evolves as

$$k_{j,t+1} = (1-\delta) k_{jt} + i_{jt}.$$

Investment costs are asymmetric:

$$\phi(i_{jt}, k_{jt}) = i_{jt} + \frac{a_P 1_{jt}^{i} + a_N (1 - 1_{jt}^{i})}{2} \left(\frac{i_{jt}}{k_{jt}} \right)^2 k_{jt},$$

with $a_N > a_P$. Cutting capital is more costly than expanding capital, which preserves the usual investment-based value-premium intuition: firms that need to shrink in bad times are especially inflexible.

3.3 Debt, retained earnings, and collateral constraints

The firm chooses next-period debt $b_{j,t+1}$. Positive values mean borrowing; negative values mean retained earnings or savings. The benchmark collateral constraint is

$$b_{j,t+1} \leq s_0 (1-\delta) k_{j,t+1}.$$

The interpretation is that debt capacity is limited by the liquidation value of capital. Since the constraint ensures full repayment, debt is risk-free in the model.

The model also allows a countercyclical liquidation-value version:

$$b_{j,t+1} \leq s_0 e^{(x_t - \bar{x}) s_1} (1-\delta) k_{j,t+1},$$

so collateral values are lower in bad times when $x_t < \bar{x}$.

Retained earnings earn a saving rate below the borrowing rate:

$$r_{st} = r_{ft} - \kappa.$$

This wedge prevents firms from simply accumulating unlimited cash and never distributing payouts.

3.4 Costly external equity

If internal funds plus new debt are insufficient to cover investment and old debt repayment, the firm issues equity:

$$e_{jt} = \max \left\{ \phi(i_{jt}, k_{jt}) + b_{jt} - \pi(k_{jt}, z_{jt}, x_t) - \frac{b_{j,t+1}}{\iota_{jt}}, 0 \right\}.$$

Equity issuance involves flotation costs:

$$\lambda(e_{jt}, k_{jt}) = \lambda_0 1^e_{jt} + \frac{\lambda_1}{2} \left(\frac{e_{jt}}{k_{jt}} \right)^2 k_{jt}.$$

A countercyclical version allows equity issuance to be more costly in bad times.

3.5 Firm value and return pricing

Equity value solves

$$v(k_{jt}, b_{jt}, z_{jt}, x_t) = \max_{i_{jt}, b_{j,t+1}} \left\{ o_{jt} + E_t[m_{t+1} v(k_{j,t+1}, b_{j,t+1}, z_{j,t+1}, x_{t+1})] \right\},$$

where effective payout is

$$o_{jt} = \pi(k_{jt}, z_{jt}, x_t) - \phi(i_{jt}, k_{jt}) + \frac{b_{j,t+1}}{\iota_{jt}} - b_{jt} - \lambda(e_{jt}, k_{jt}).$$

Expected returns satisfy the usual beta-pricing condition:

$$E_t[r_{j,t+1}] - r_{ft} = \beta_{jt} \zeta_{mt}, \qquad \beta_{jt} = -\frac{\mathrm{Cov}_t(r_{j,t+1}, m_{t+1})}{\mathrm{Var}_t(m_{t+1})}.$$

4 The shadow price of new debt

The key state-dependent measure of financial constraints is the shadow price of new debt, denoted ν_{jt}. It is the Lagrange multiplier on the collateral constraint. In the model,

$$\nu_{jt} = \frac{1}{r_{ft}} \lambda_e(e_{jt}, k_{jt}) 1^e_{jt} - E_t \left[m_{t+1} \lambda_e(e_{j,t+1}, k_{j,t+1}) 1^e_{j,t+1} \right].$$

Interpretation. One more unit of debt relaxes the need to issue costly equity today, but it also creates a repayment burden tomorrow that may force costly equity issuance later. The shadow price is high when the marginal value of relaxing the current financing constraint exceeds the expected future cost of repayment.

The model implies that the shadow price is higher for firms with small capital stocks, low firm-specific productivity, and high current debt. These firms have higher financing deficits: desired investment is high relative to internal funds and borrowing capacity.

5 Simulation design and calibration

The paper solves the model at monthly frequency by value function iteration. The benchmark calibration uses a production curvature $\alpha = 0.65$, monthly depreciation $\delta = 0.01$, persistent aggregate productivity, asymmetric adjustment costs, persistent firm-level productivity, collateral liquidation value $s_0 = 0.85$, fixed equity flotation cost $\lambda_0 = 0.08$, convex flotation cost $\lambda_1 = 0.025$, and a monthly wedge between borrowing and saving rates of 0.50%/12.

For quantitative experiments, the authors simulate 100 artificial panels. Each panel contains 3,000 firms and 480 monthly observations. They then apply standard empirical asset-pricing procedures, including value-weighted portfolio sorts and Fama-MacBeth cross-sectional regressions, to the simulated data.

6 Main quantitative findings

6.1 Financial constraints and average returns

In one-way sorts on the shadow price of new debt, more constrained firms earn higher average stock returns. Under the benchmark calibration, the lowest-shadow-price portfolio earns 0.33% per month, while the highest-shadow-price portfolio earns 0.67% per month, giving a high-minus-low spread of 0.34% per month.

After controlling for size through two-way sorts, the financial-constraints factor is much smaller and statistically weak. This result is consistent with the empirical evidence of Lamont, Polk, and Saa-Requejo (2001) and Whited and Wu (2006). The interpretation is not that constraints are irrelevant, but that size and book-to-market contain overlapping information about the same underlying firm state variables.

6.2 Why constraints raise risk

Financially constrained firms cannot fully use investment to smooth payouts after aggregate shocks. When a positive aggregate shock occurs, unconstrained firms raise investment, partially absorbing the shock and smoothing dividends. Constrained firms cannot adjust investment as much, so their dividends covary more strongly with aggregate productivity. This raises beta and expected returns.

6.3 Determinants of financial constraints

The simulations reproduce the signs of many empirical constraints predictors. Firms are more constrained when they have low cash flow-to-assets, high debt-to-assets, small asset size, low sales growth, low dividend payouts, low cash holdings, and high Tobin's Q. These characteristics proxy for the firm's financing deficit and its need for external funds.

6.4 KZ versus WW indexes

Both the Kaplan-Zingales and Whited-Wu indexes are positively correlated with the true model-implied shadow price of new debt. However, both are noisy. In most regressions, the R^2 values are below 10%, meaning that the indexes leave most variation in true financial constraints unexplained. The Whited-Wu index generally performs better than the Kaplan-Zingales index in the model.

6.5 Leverage and returns

The model predicts a convex relation between market leverage and expected stock returns. The textbook mechanism says that holding asset beta fixed, higher leverage mechanically raises equity beta. This paper adds a second channel: higher leverage also raises asset beta because debt repayment reduces investment flexibility. Thus leverage affects expected returns both mechanically and through real investment inflexibility.

7 Identification and empirical design

This is primarily a structural asset-pricing paper rather than a reduced-form empirical identification paper. Its identifying logic comes from a fully specified model in which the true financial constraint is observable in simulation. The authors then ask whether empirical procedures applied to simulated panels reproduce patterns seen in real data.

The paper uses three main empirical-style designs inside the model:

- **Portfolio sorts on the shadow price.** These show the unconditional constraints-return relation.
- **Two-way sorts on size and constraints.** These test whether the relation survives after controlling for market capitalization.
- **Fama-MacBeth regressions.** These quantify marginal effects of the shadow price, size, book-to-market, leverage, and asset beta.

The interpretation is model-based. If the simulated model can replicate weak or conflicting empirical constraints-return evidence while generating a positive true constraints-risk relation, then empirical constraints indexes may be too noisy to reveal the underlying mechanism clearly.

8 Extracted figures and tables from the paper

8.1 Table I: Benchmark Parameter Values

Read. Table I lists the benchmark monthly calibration. Key parameters include $\alpha = 0.65$, $\delta = 0.01$, $a_P = 15$, $a_N = 150$, $s_0 = 0.85$, $\lambda_0 = 0.08$, and $\lambda_1 = 0.025$.

Economic message. The calibration combines standard investment adjustment costs, collateral constraints, and equity issuance costs. The asymmetric adjustment costs and financial frictions are both necessary for the inflexibility channel.

8.2 Figure 1: The Market Equity-to-Capital and Optimal Investment-to-Capital Ratios against Underlying State Variables, the Benchmark Parametrization

Read. Figure 1 plots equity value scaled by capital and investment scaled by capital as functions of capital, debt, aggregate productivity, and firm-specific productivity.

Economic message. Small, productive firms have high valuation ratios and strong investment incentives. High current debt lowers equity value and depresses investment, showing how debt burdens affect real decisions.

Benchmark Parameter Values

This table lists the benchmark parameter values used to solve and simulate our model with the collateral constraints.

Parameter	Value	Description
α	0.65	Curvature in the production function
δ	0.01	Monthly rate of capital depreciation
ρ_x	$0.95^{1/3}$	Persistence coefficient of aggregate productivity
σ_x	0.007/3	Conditional volatility of aggregate productivity
η	0.994	Time preference coefficient
γ_0	50	Constant price of risk parameter
γ_1	−1,000	Time-varying price of risk parameter
a_P	15	Adjustment cost parameter when investment is positive
a_N	150	Adjustment cost parameter when investment is negative
ρ_z	0.96	Persistence coefficient of firm-specific productivity
σ_z	0.10	Conditional volatility of firm-specific productivity
f	0.015	Fixed costs of production
s_0	0.85	Liquation value per unit of capital net of (acyclical) bankruptcy cost
s_1	0	Countercyclical liquidation cost parameter
λ_0	0.08	Fixed flotation cost parameter
λ_1	0.025	Convex (acyclical) flotation cost parameter
λ_2	0	Countercyclical flotation cost parameter
κ	0.50%/12	Monthly wedge between the borrowing and saving rates of interest

Table I: Benchmark Parameter Values

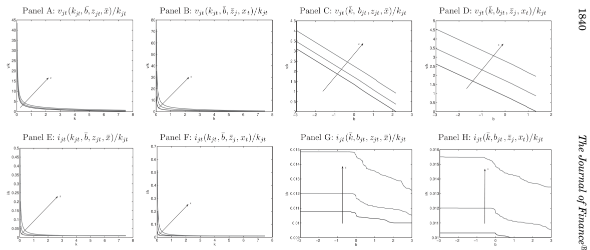


Figure 1. The market equity-to-capital and optimal investment-to-capital ratios against underlying state variables; the benchmark parametrization. We plot the ratio of market value of equity to capital (v_{jt}/k_{jt}, Panels A–D) and the optimal investment-to-capital ratio (i_{jt}/k_{jt}, Panels E–H) as functions of underlying state variables. Panels A and E plot the variables as functions of current-period capital, k_{jt}, and firm-specific productivity, z_{jt}, while fixing aggregate productivity x_t and current-period debt b_{jt} at their respective long-run average levels ($\bar{x}$ and $\bar{b}$, respectively; $\bar{b}$ is determined in model simulations). Panels A and E have a class of curves, corresponding to different values of z_{jt}, and the arrow in each panel indicates the direction along which z_{jt} increases. Panels B and F plot the variables as functions of current-period capital, k_{jt}, and aggregate productivity, x_t, while fixing firm-specific productivity z_{jt} and current-period debt b_{jt} at their respective long-run average levels (zero and $\bar{b}$, respectively). Panels B and F have a class of curves, corresponding to different values of x_t, and the arrow in each panel indicates the direction along which x_t increases. Panels C and G plot the variables as functions of current-period debt, b_{jt}, and firm-specific productivity, z_{jt}, while fixing aggregate productivity x_t and capital stock k_{jt} at their respective long-run average levels ($\bar{x}$ and $\bar{k}$, respectively; $\bar{k}$ is determined in model simulations). Panels C and G have a class of curves, corresponding to different values of z_{jt}, and the arrow in each panel indicates the direction along which z_{jt} increases. Panels D and H plot the variables as functions of current-period debt, b_{jt}, and aggregate productivity x_t, while fixing firm-specific productivity z_{jt} and capital stock k_{jt} at their long-run average levels (zero and $\bar{k}$, respectively). Panels D and H have a class of curves, corresponding to different values of x_t, and the arrow in each panel indicates the direction along which x_t increases.

Figure 1: The Market Equity-to-Capital and Optimal Investment-to-Capital Ratios against Underlying State Variables, the Benchmark Parametrization

8.3 Figure 2: The Optimal Debt-to-Capital Ratio and the Shadow Price of New Debt against Underlying State Variables, the Benchmark Parametrization

Read. Figure 2 shows optimal next-period debt-to-capital and the shadow price of new debt as functions of the state variables.

Economic message. The shadow price is high for firms with small capital, low productivity, and high existing debt. These firms have the largest financing deficits and are therefore the most constrained.

8.4 Figure 3: Risk and Expected Excess Returns against Underlying State Variables, the Benchmark Parametrization

Read. Figure 3 plots beta and expected excess returns. Risk and expected returns are higher for firms with small capital, low productivity, and high current debt.

Economic message. The same state variables that make firms financially constrained also make them riskier. This visual is the bridge from corporate financing frictions to the cross-section of expected returns.

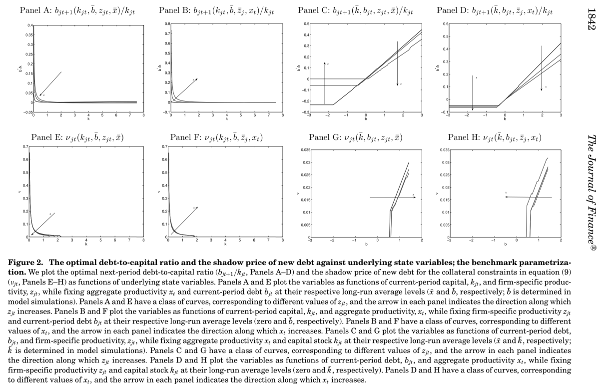


Figure 2: The Optimal Debt-to-Capital Ratio and the Shadow Price of New Debt against Underlying State Variables, the Benchmark Parametrization

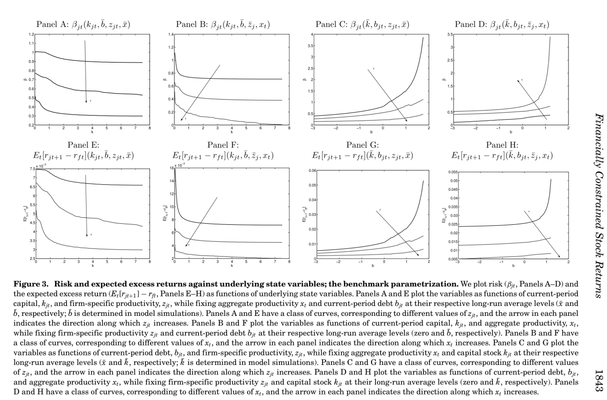


Figure 3: Risk and Expected Excess Returns against Underlying State Variables, the Benchmark Parametrization

8.5 Table II: Average Monthly Percent Stock Returns for Portfolios Based on One-Way Sorts on the Shadow Price of New Debt in Model Simulations

Read. Table II sorts firms into deciles by the shadow price of new debt. Under the benchmark calibration, returns rise monotonically from 0.33% to 0.67% per month, giving a high-minus-low spread of 0.34% per month.

Economic message. The true model-implied financial constraint is positively priced. More constrained firms are riskier because they are less able to smooth payouts through investment.

8.6 Table III: Average Monthly Percentage Excess Returns for Portfolios Sorted on the Shadow Price of New Debt and Market Capitalization in Model Simulations

Read. Table III performs two-way sorts on size and the shadow price. The benchmark FC portfolio earns 0.12% per month with a t-statistic of 1.11.

Economic message. Once size is controlled for, the constraints-return relation becomes weak. This helps reconcile the model's positive true constraints-risk relation with weak empirical evidence after size controls.

Table II

Average Monthly Percent Stock Returns for Portfolios Based on One-Way Sorts on the Shadow Price of New Debt in Model Simulations

This table reports average stock returns of 10 value-weighted portfolios from one-way sorts on the shadow price of new debt on the collateral constraints. We report average returns in percent per month for each portfolio as well as the average high-minus-low portfolios and its t-statistics. We sort all firms based on their shadow prices of new debt at the beginning of each year and then hold the portfolios for the whole year. For each model, we simulate 100 artificial panels, each of which has 3,000 firms and 480 monthly observations, and we then report the across-simulation average results. We report the simulation results from the model using the benchmark parameters reported in Table I (Benchmark parametrization). We also report the results from four comparative static experiments ($s_0 = 0.70$, $\lambda_0 = 0.02$, $s_1 > 0$, and $\lambda_2 > 0$).

Low	2	3	4	5	6	7	8	9	High	*FC*	t_{FC}
Panel A: Benchmark Parametrization											
0.33	0.37	0.43	0.44	0.46	0.50	0.54	0.57	0.61	0.67	0.34	3.54
Panel B: High Liquidation Costs, $s_0 = 0.70$											
0.14	0.14	0.15	0.15	0.16	0.16	0.18	0.18	0.18	0.19	0.05	2.23
Panel C: Low Equity Flotation Costs, $\lambda_0 = 0.02$											
0.37	0.40	0.45	0.50	0.52	0.53	0.58	0.61	0.62	0.62	0.25	4.63
Panel D: Countercyclical Liquidation Costs, $s_1 > 0$											
0.11	0.12	0.14	0.15	0.16	0.16	0.17	0.18	0.18	0.20	0.09	2.70
Panel E: Countercyclical Equity Flotation Costs, $\lambda_2 > 0$											
0.34	0.40	0.44	0.49	0.53	0.60	0.63	0.70	0.75	0.79	0.45	4.53

Table II: Average Monthly Percent Stock Returns for Portfolios Based on One-Way Sorts on the Shadow Price of New Debt in Model Simulations

Table III

Average Monthly Percentage Excess Returns for Portfolios Sorted on the Shadow Price of New Debt and Market Capitalization in Model Simulations

We report average returns in monthly percent in excess of the risk-free rate for nine value-weighted portfolios sorted on market capitalization and the shadow price of new debt in model simulations. The rankings are performed annually and independently such that each portfolio contains firms in both a given size category and a given financial constraints category. Following Lamont, Polk, and Saá-Requejo (2001) and Whited and Wu (2006), we define small-caps (S) as firms that are in the bottom 40%, mid-caps (M) are firms in the middle 20%, and large-caps (B) are firms in the top 40% of the sample sorted on market capitalization. Similarly, low-, middle-, and high-shadow-price portfolios consist of firms in the bottom 40% (L), middle 20% (M), and top 40% (H) of the sample sorted on the shadow price of new debt, respectively. We also define the average high-*FC* portfolio as $HIGH\,FC \equiv (BH + MH + SH)/3$, and average low-*FC* portfolio as $LOW\,FC \equiv (BL + ML + SL)/3$, and the financial constraints factor as $FC \equiv HIGH\,FC - LOW\,FC$. For each model we simulate 100 artificial panels, each of which has 3,000 firms and 480 monthly observations, and we then report the cross-simulation average results. The column denoted "Benchmark" reports the simulation results from the model using the benchmark parameter values in Table I. We also report results from four comparative static experiments. The column denoted "$s_0 = 0.70$" reports the results using the benchmark parameters except that the liquidation value per unit of capital net of liquidation costs, s_0, is reset to be 0.70. The column denoted "$\lambda_0 = 0.02$" reports the results using the benchmark parameters except that the fixed flotation cost of equity, λ_0, is reset to be 0.02. The column denoted "$s_1 > 0$" reports the results from the countercyclical liquidation costs model with $s_1 > 0$ (see equation (10)). The column denoted "$\lambda_2 > 0$" reports the results from the countercyclical equity flotation costs model with $\lambda_2 > 0$ (see equation (15)). The last two columns report those from Table I of Lamont et al. and from Table 4 of Whited and Wu, respectively.

		Benchmark	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_1 > 0$	$\lambda_2 > 0$	Lamont et al. (2001)	Whited and Wu (2006)
Small-caps								
Low *FC*	*SL*	0.61	0.25	0.68	0.28	0.68	0.45	0.89
Middle *FC*	*SM*	0.64	0.31	0.75	0.34	0.84	0.67	0.66
High *FC*	*SH*	0.75	0.40	0.88	0.38	0.91	0.38	0.83
Mid-caps								
Low *FC*	*ML*	0.45	0.14	0.56	0.16	0.56	0.37	0.65
Middle *FC*	*MM*	0.50	0.16	0.60	0.21	0.74	0.56	0.81
High *FC*	*MH*	0.59	0.16	0.65	0.25	0.84	0.26	0.74
Large-caps								
Low *FC*	*BL*	0.21	0.11	0.37	0.14	0.42	0.47	0.71
Middle *FC*	*BM*	0.30	0.08	0.41	0.15	0.51	0.53	0.96
High *FC*	*BH*	0.37	0.09	0.50	0.18	0.59	0.25	1.23
HIGH FC		0.51	0.22	0.63	0.26	0.74	0.30	0.93
LOW FC		0.39	0.06	0.51	0.16	0.67	0.43	0.75
FC		0.12	0.16	0.12	0.10	0.07	−0.13	0.18
t_{FC}		1.11	0.56	0.98	0.89	0.63	−1.17	0.95

Table III: Average Monthly Percentage Excess Returns for Portfolios Sorted on the Shadow Price of New Debt and Market Capitalization in Model Simulations

8.7 Table IV: Fama-MacBeth (1973) Monthly Cross-Sectional Regressions of Percentage Stock Returns on the Shadow Price of New Debt, Size, and Book-to-Market in Model Simulations

Read. Table IV shows that the shadow price predicts returns in univariate regressions, but its slope becomes insignificant or negative when size and book-to-market are included.

Economic message. Financial constraints, size, and book-to-market are all endogenous functions of the state variables. The shadow price contains risk information, but some of that information overlaps with size and book-to-market.

8.8 Table V: Cross-Sectional Determinants of the Shadow Price of New Debt in Model Simulations

Read. Table V regresses the shadow price on characteristics used in the Whited-Wu and Kaplan-Zingales indexes. The model generally matches the empirical signs: constrained firms have lower cash flow, higher debt, smaller size, lower sales growth, lower dividends, lower cash, and higher Tobin's Q.

Economic message. The model rationalizes why standard observables proxy for financial constraints. However, these observables are only partial and noisy measures of the true shadow price.

Table IV

Fama-MacBeth (1973) Monthly Cross-sectional Regressions of Percentage Stock Returns on the Shadow Price of New Debt, Size, and Book-to-Market in Model Simulations

We report the Fama-MacBeth (1973) monthly cross-sectional regressions of stock returns, r_{jt+1}, on the shadow price of new debt, size, and book-to-market, all of which are measured at the beginning of month t. ν_{jt} denotes the shadow price, ln(ME) is the logarithm of the market value of equity, measured as $\ln(v_{jt} - o_{jt})$ in the model, and ln(BE/ME) is the logarithm of the book-to-market equity ratio, measured as $\ln((k_{jt} - b_{jt})/(v_{jt} - o_{jt}))$ in the model. We simulate 100 artificial panels, each of which has 3,000 firms and 480 monthly observations, and then report the across-simulation average slopes and t-statistics. We report the results from the model with the benchmark parametrization in Table I (Benchmark). We also report four comparative static experiments ($s_0 = 0.70$, $\lambda_0 = 0.02$, $s_1 > 0$, and $\lambda_2 > 0$).

Benchmark			High Liquidation Cost ($s_0 = 0.70$)		
ν_{jt}	ln (ME)	ln (B/M)	ν_{jt}	ln (ME)	ln (B/M)
1.69			1.17		
(3.47)			(2.10)		
2.52	−1.96	3.67	0.63	−3.01	2.03
(0.79)	(−2.55)	(3.07)	(1.59)	(−2.26)	(1.03)

Low Fixed Flotation Costs ($\lambda_0 = 0.02$)			Countercyclical Liquidation Costs ($s_1 > 0$)			Countercyclical Flotation Costs ($\lambda_2 > 0$)		
ν_{jt}	ln (ME)	ln (B/M)	ν_{jt}	ln (ME)	ln (B/M)	ν_{jt}	ln (ME)	ln (B/M)

Table IV: Fama-MacBeth (1973) Monthly Cross-Sectional Regressions of Percentage Stock Returns on the Shadow Price of New Debt, Size, and Book-to-Market in Model Simulations

Table V

Cross-sectional Determinants of the Shadow Price of New Debt in Model Simulations

We report Fama-MacBeth (1973) monthly cross-sectional regressions of the shadow price of new debt on firm characteristics. Panel A reports the regression similar to that in column 4 of Table I in Whited and Wu (2006): $\nu_{jt} = b_0 + b_1\, CF_{jt} + b_2\, TLTD_{jt} + b_3\, LNTA_{jt} + b_4\, SG_{jt} + b_5\, DIVPOS_{jt} + \epsilon_{jt}$, where ν_{jt} is the shadow price for firm j in month t. CF_{jt} is the cash flow-to-asset ratio, $TLTD_{jt}$ is the total debt-to-assets ratio, $LNTA_{jt}$ is the logarithm of the total assets, SG_{jt} is the firm-level sales growth, and $DIVPOS_{jt}$ is a dummy variable that equals one if firm j has paid dividends during month t, and equals zero otherwise. Panel B reports the regression similar to that reported in Table 9 in Lamont, Polk, and Saá-Requejo (2001), the regression that is in turn based on Kaplan and Zingales (1997): $\nu_{jt} = b_0 + b_1\, CF_{jt} + b_2\, TLTD_{jt} + b_3\, TDIV_{jt} + b_4\, CASH_{jt} + b_5\, Q_{jt} + \epsilon_{jt}$, where CF_{jt} is the cash flow-to-asset ratio, $TLTD_{jt}$ is the total debt-to-assets ratio, $TDIV_{jt}$ is the dividend-to-assets ratio, $CASH_{jt}$ is the ratio of total amount of liquid assets or cash divided by assets, and Q_{jt} is Tobin's Q. The t-statistics are in parentheses. We report quantitative results from the benchmark parametrization reported in Table I and four comparative static experiments: (i) the high liquidation costs case with $s_0 = 0.70$; (ii) the low fixed flotation costs case with $\lambda_0 = 0.02$; (iii) the countercyclical liquidation costs case with $s_1 > 0$; and (iv) the countercyclical equity flotation costs case with $\lambda_2 > 0$. For each case, we simulate 100 artificial panels, each of which has 3,000 firms and 480 months, and report across-simulation average results. We compare our results to those of Lamont et al. and Whited and Wu (reported under "Data").

Panel A: $\nu_{jt} = b_0 + b_1\, CF_{jt} + b_2\, TLTD_{jt} + b_3\, LNTA_{jt} + b_4\, SG_{jt} + b_5\, DIVPOS_{jt} + \epsilon_{jt}$ from Whited and Wu (2006)

CF		*TLTD*		*LNTA*		*SG*		*DIVPOS*	
Data	Benchmark	Data	Benchmark	Data	Benchmark	Data	Benchmark	Data	Benchmark
−0.09	−0.25	0.02	0.14	−0.04	−0.11	−0.04	−0.02	−0.06	−0.37
(−2.94)	(−10.24)	(1.91)	(4.59)	(−1.91)	(−10.53)	(−1.52)	(−5.88)	(−2.14)	(−3.67)
$s = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$
−0.08	−0.33	0.03	0.02	−0.03	−0.03	−0.00	−0.05	−0.07	−0.13
(−2.36)	(−2.81)	(3.84)	(5.88)	(−6.11)	(−8.29)	(−3.46)	(−1.50)	(−2.23)	(−2.83)
$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$
−0.12	−0.03	0.03	0.01	−0.01	−0.01	−0.00	−0.00	−0.12	−0.03
(−2.78)	(−2.55)	(8.60)	(2.22)	(−0.89)	(−2.83)	(−0.42)	(−0.31)	(−1.79)	(−1.57)

Panel B: $\nu_{jt} = b_0 + b_1\, CF_{jt} + b_2\, TLTD_{jt} + b_3\, TDIV_{jt} + b_4\, CASH_{jt} + b_5\, Q_{jt} + \epsilon_{jt}$ from Kaplan and Zingales (1997)

CF		*TLTD*		*TDIV*		*CASH*		*Q*	
Data	Benchmark	Data	Benchmark	Data	Benchmark	Data	Benchmark	Data	Benchmark
−1.00	−2.50	3.14	1.78	−39.37	−3.61	−1.32	−0.10	0.28	0.20
(−4.28)	(−5.60)	(6.99)	(9.23)	(−6.46)	(−9.95)	(−4.55)	(−7.30)	(3.63)	(7.47)
$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$	$s_0 = 0.70$	$\lambda_0 = 0.02$
−0.35	−1.36	1.33	0.23	−3.46	−1.36	−0.00	−0.24	1.70	0.12
(−3.08)	(−2.93)	(9.54)	(5.77)	(−2.07)	(−2.91)	(−7.28)	(−8.10)	(7.27)	(5.70)
$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$	$s_1 > 0$	$\lambda_2 > 0$
−1.80	−1.88	1.25	1.59	−1.77	−1.19	−0.06	−0.00	0.06	0.01
(−2.66)	(−2.24)	(8.60)	(2.23)	(−2.60)	(−1.27)	(−2.83)	(−2.95)	(7.97)	(3.04)

Table V: Cross-Sectional Determinants of the Shadow Price of New Debt in Model Simulations

8.9 Table VI: Fama-MacBeth (1973) Cross-Sectional Regressions of the Shadow Price of New Debt on the Kaplan and Zingales (1997) and the Whited and Wu (2006) Indexes of Financial Constraints

Read. Table VI compares the KZ and WW indexes as proxies for the true shadow price. Both indexes are positively related to the shadow price, but the R^2 values are low. WW generally has a larger slope and performs better.

Economic message. Empirical financial-constraint indexes are informative but very noisy. This is a central reason why empirical constraints-return tests may generate weak or conflicting results.

8.10 Figure 4: The Relation between Equity Risk and Market Leverage, the Benchmark Parametrization

Read. Figure 4 plots equity beta against market leverage, holding aggregate productivity and capital fixed while varying firm-specific productivity.

Economic message. The leverage-risk relation is convex, especially for low-productivity firms. Leverage increases risk not only mechanically through capital structure, but also by tightening financing constraints and reducing real investment flexibility.

variation in the true shadow price unexplained. From the relative magnitude of the slopes, the *WW* index appears to do a better job than the *KZ* index as a

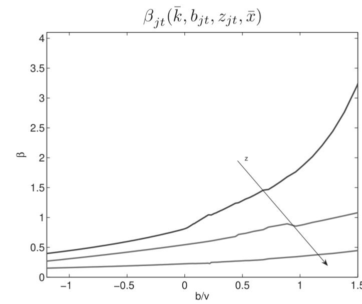
Table VI

Fama-MacBeth (1973) Cross-sectional Regressions of the Shadow Price of New Debt on the Kaplan and Zingales (1997) and the Whited and Wu (2006) Indexes of Financial Constraints

Using simulated panels, we report the Fama-MacBeth (1973) monthly cross-sectional regressions of the shadow price of new debt, v_{jt}, on the Kaplan and Zingles (1997, *KZ*) index and the Whited and Wu (2006, *WW*) index of financial constraints, both separately and jointly. To make the magnitudes of their slopes comparable, we standardize both indexes by dividing their demeaned values by their respective standard deviations before using them in the cross-sectional regressions. We simulate 100 artificial panels, each of which has 3,000 firms and 480 monthly observations, and then report across-simulation average Fama-MacBeth slopes, *t*-statistics (in parentheses), and average cross-sectional R^2s. We also report the average cross-sectional correlation between the *KZ* index and the *WW* index. Panel A reports the quantitative results from the benchmark parametrization with the parameter values in Table I. The remaining panels report four comparative static experiments: the high liquidation costs case with $s_0 = 0.70$ (Panel B), the low fixed flotation costs case with $\lambda_0 = 0.02$, the countercyclical liquidation costs case with $s_1 > 0$, and the countercyclical flotation costs case with $\lambda_2 > 0$.

KZ	R^2	*WW*	R^2	*KZ*	*WW*	R^2	Corr(*KZ*, *WW*)
Panel A: Benchmark Parametrization							
0.017	0.05	0.108	0.10	0.014	0.108	0.12	0.50
(5.28)		(11.47)		(2.97)	(11.53)		
Panel B: High Liquidation Costs, $s_0 = 0.70$							
0.010	0.04	0.080	0.10	0.010	0.063	0.10	0.56
(7.22)		(13.30)		(7.28)	(4.51)		
Panel C: Low Fixed Flotation Costs, $\lambda_2 = 0.02$							
0.021	0.07	0.151	0.09	0.011	0.120	0.09	0.47

Table VI: Fama-MacBeth (1973) Cross-Sectional Regressions of the Shadow Price of New Debt on the Kaplan and Zingales (1997) and the Whited and Wu (2006) Indexes of Financial Constraints



igure 4. The relation between equity risk and market leverage, the benchmar arametrization. We plot risk (β_{jt}) against market leverage (b_{jt}/v_{jt}), while fixing aggregate pr ıctivity x_t and capital stock k_{jt} at their long-run average levels ($\bar{x}$ and $\bar{k}$, respectively). $\bar{k}$, dete ined in simulations, is around one. The panel has a class of curves that correspond to differe ılues of z_{jt}. The arrow indicates the direction along which z_{jt} increases.

here β_{jt} is the equity beta as defined in equation (21). Table VII shows tha hen used alone, market leverage significantly covaries with future stock re

Figure 4: The Relation between Equity Risk and Market Leverage, the Benchmark Parametrization

8.11 Table VII: Fama-MacBeth (1973) Monthly Cross-Sectional Regressions of Stock Returns on Market Leverage, with and without Controlling for Asset Beta

Read. Table VII regresses stock returns on market leverage, with and without controlling for asset beta. Market leverage alone predicts returns, but its coefficient becomes weaker and insignificant once asset beta is included, while asset beta remains positive and significant.

Economic message. The leverage-return relation in the model is driven largely by leverage's effect on real asset risk through financial inflexibility. This goes beyond the simple textbook mechanical leverage channel.

Table VII

Fama-MacBeth (1973) Monthly Cross-sectional Regressions of Stock Returns on Market Leverage, with and without Controlling for Asset Beta

Using simulated panels, we report the Fama-MacBeth (1973) monthly cross-sectional regressions of the stock returns, r_{jt+1}, on market leverage, b_{jt}/v_{jt}, with and without controlling for the asset beta, β^A_{jt}. β^A_{jt} is defined as $\beta_{jt}v_{jt}/(b_{jt}+v_{jt})$, where β_{jt} is the equity beta. Appendix B details the numerical calculations of β_{jt}. We simulate 100 artificial panels, each of which has 3,000 firms and 480 monthly observations, and then report across-simulation average Fama-MacBeth slopes, t-statistics (in parentheses), and average cross-sectional R^2s. Panel A reports the quantitative results from the benchmark parametrization with the parameter values in Table I. The remaining panels report four comparative static experiments: the high liquidation costs case with $s_0 = 0.70$ (Panel B), the low fixed flotation costs case with $\lambda_0 = 0.02$ (Panel C), the countercyclical liquidation costs case with $s_1 > 0$ (Panel D), and the countercyclical flotation costs case with $\lambda_2 > 0$ (Panel E).

b_{jt}/v_{jt}	R^2	b_{jt}/v_{jt}	β^A_{jt}	R^2
Panel A: Benchmark Parametrization				
1.093	0.28	1.051	0.0049	0.39
(5.15)		(1.26)	(3.31)	
Panel B: High Liquidation Costs, $s_0 = 0.70$				
3.617	0.14	2.456	0.0048	0.15
(7.22)		(0.46)	(10.29)	
Panel C: Low Fixed Flotation Costs, $\lambda_2 = 0.02$				
3.490	0.09	3.083	0.0062	0.12
(4.37)		(0.82)	(10.40)	
Panel D: Countercyclical Liquidation Costs, $s_1 > 0$				
3.827	0.14	2.079	0.0031	0.15
(6.58)		(0.78)	(9.60)	
Panel E: Countercyclical Flotation Costs, $\lambda_2 > 0$				
3.062	0.08	1.032	0.0046	0.09
(8.55)		(1.74)	(9.63)	

The interplay between asset pricing and corporate finance is likely to remain

Table VII: Fama-MacBeth (1973) Monthly Cross-Sectional Regressions of Stock Returns on Market Leverage, with and without Controlling for Asset Beta

9 How to read the paper for research use

9.1 For asset pricing

The paper is useful because it shows how corporate financing frictions can be priced even without default risk. The priced object is not distress, but the firm's inability to adjust investment flexibly after aggregate shocks. This makes the paper closely related to investment-based asset pricing, value premium models, and production-based interpretations of expected returns.

9.2 For corporate finance

The paper turns financial constraints from an empirical index problem into a model object: the shadow price of new debt. This clarifies why constraints correlate with low cash flow, high debt, low size, low dividends, low cash, and high Tobin's Q, but also why no single empirical index is likely to be sufficient.

9.3 For empirical work

The paper suggests caution when using the KZ or WW indexes as direct measures of financial constraints. They are positively correlated with the true shadow price in the model, but they explain little of its variation. Empirical tests that use these indexes may be affected by substantial measurement error.

10 Limitations and open questions

- **No defaultable debt.** Debt is risk-free because the collateral constraint enforces full repayment. This makes the model well suited for financial constraints, but not for financial distress.
- **Partial-equilibrium discount factor.** The stochastic discount factor is specified exogenously. A fully general-equilibrium model might change the quantitative results.
- **Shadow price is unobservable.** The model's main constraint measure is clean internally, but empirical work must rely on noisy proxies.
- **No rich debt maturity or renegotiation.** Debt is one-period, and the model abstracts from defaultable bonds, debt maturity structure, covenants, renegotiation, and agency conflicts.
- **Calibration matters.** The return spreads vary across liquidation cost and flotation cost parameterizations, so the quantitative magnitude is model-dependent.

11 Short summary

Livdan, Sapriza, and Zhang (2009) show that financially constrained firms should be riskier and earn higher expected returns because collateral constraints prevent them from fully using investment to smooth payouts after aggregate shocks. The model's true constraint measure is the shadow price of new debt. In one-way simulated portfolio sorts, high-shadow-price firms outperform low-shadow-price firms. After controlling for size and book-to-market, the relation weakens because these variables share information about the same underlying state variables. The paper also shows that common empirical constraints indexes are noisy proxies for true constraints and that leverage raises expected returns through both a standard equity-risk channel and a real investment-inflexibility channel.

12 Conclusion

The paper's central contribution is to connect financial constraints, corporate investment, debt dynamics, and expected stock returns in one model. Its key insight is that constraints matter for asset pricing because they reduce flexibility. A constrained firm cannot fully adjust investment to absorb aggregate shocks, so its payouts are more cyclical and its stock is riskier. This mechanism offers a structural interpretation of financial constraints risk, explains why empirical evidence can be weak after size controls, and provides a new view of the leverage-return relation.